

Yunsoo Ha

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ACADEMIC APPOINTMENTS

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| • Cornell University
Postdoctoral Associate, School of Civil and Environmental Engineering | 2025 – Present
USA |
| • National Laboratory of the Rockies
Postdoctoral Researcher, Computational Science Center | 2024 – 2025
USA |

EDUCATION

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|--|----------------------------|
| • Ph.D. in Industrial and Systems Engineering
North Carolina State University
◦ Advisor: Sara Shashaani
◦ Committee: Yunan Liu, Reha Uzsoy, Quoc Tran-Dinh, Naihuan Jing
◦ Dissertation: <i>Expediting Stochastic Derivative-Free Optimization</i>
🏆 ISE Distinguished Dissertation Award of the 2024 CA Anderson Awards
🏆 Second Place Winner of the 2025 Pritsker Doctoral Dissertation Award by IISE | 2018 – 2024
USA |
| • Master of Operations Research
North Carolina State University | 2018 – 2021
USA |
| • M.S. in Logistics, School of Air Transport, Transportation, and Logistics
Korea Aerospace University | 2015 – 2017
South Korea |
| • B.S. in Logistics, School of Air Transport, Transportation, and Logistics
Korea Aerospace University | 2010 – 2015
South Korea |

RESEARCH INTERESTS

Methods: Simulation Optimization, Stochastic and Derivative-Free Optimization, Monte Carlo Methods
Directions: Decision-Focused Learning, Digital Twin Optimization, Quantum-Classical Optimization

PUBLICATIONS

Peer-Reviewed Journal Articles

- [1] **Adaptive Sampling Bi-Fidelity Stochastic Trust Region Method for Derivative-Free Stochastic Optimization.**
Yunsoo Ha, Juliane Mueller.
Mathematical Programming Computation (2026).
- [2] **Complexity of Zeroth- and First-Order Stochastic Trust-Region Algorithms.**
Yunsoo Ha, Raghu Pasupathy, Sara Shashaani.
SIAM Journal on Optimization (2025).
- [3] **Two-Stage Estimation and Variance Modeling for Latency-Constrained Variational Quantum Algorithms.**
Yunsoo Ha, Sara Shashaani, Matt Menickelly.
INFORMS Journal on Computing (2025).
- [4] **Iteration Complexity and Finite-Time Efficiency of Adaptive Sampling Trust-Region Methods for Stochastic Derivative-Free Optimization.**
Yunsoo Ha, Sara Shashaani.
IIE Transactions (2025).
- [5] **Latency Considerations for Stochastic Optimizers in Variational Quantum Algorithms.**
Matt Menickelly, Yunsoo Ha, Matthew Otten.
Quantum (2023).
- [6] **A Decision Model to Determine the Number of Shuttles in a Tier-to-Tier SBS/RS.**
Yunsoo Ha, Junjae Chae.
International Journal of Production Research (2019).

[7] **Free Balancing for a Shuttle-Based Storage and Retrieval System.**

Yunsoo Ha, Junjae Chae.

Simulation Modelling Practice and Theory (2018).

Manuscripts Under Review

[1] **Computing Equilibria in Simulation-Based Insurance Markets with Discontinuous Demand.**

Yunsoo Ha, Linda Nozick.

Under review at *European Journal of Operational Research*.

[2] **Adaptive Regularization within Trust Region Methods for Stochastic Nonconvex Optimization.**

Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.

Under review at *Mathematical Programming*.

🏆 *Third Place Poster Award, 2026 INFORMS Simulation Society Research Workshop*

Working Papers

[1] **Multilevel Monte Carlo Trust-Region Method with Adaptive Sampling for Nested Simulation Optimization.**

Yunsoo Ha, Hyunwoo Shin.

[2] **Stochastic Derivative-Free Optimization with Adaptive Regularization and Adaptive Sampling Trust Region Methods.**

Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.

[3] **Decision-Focused Input Modeling for Queueing Systems.**

Yunsoo Ha.

Peer-Reviewed Conference Proceedings

[1] **Multi-Fidelity Stochastic Trust Region Method with Adaptive Sampling.**

Yunsoo Ha, Julianne Mueller.

Winter Simulation Conference 2025.

[2] **Towards Greener Stochastic Derivative-Free Optimization with Trust Regions and Adaptive Sampling.**

Yunsoo Ha, Sara Shashaani.

Winter Simulation Conference 2023.

[3] **Improved Complexity of Trust-Region Optimization for Zeroth-Order Stochastic Oracles With Adaptive Sampling.**

Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.

Winter Simulation Conference 2021.

HONORS AND AWARDS

• Third Place Poster Presentation Award <i>INFORMS Simulation Society Research Workshop</i>	2026
• Second Place Winner of the 2025 Pritsker Doctoral Dissertation Award <i>Institute of Industrial and Systems Engineers</i>	2025
• Outstanding Reviewer Award <i>Winter Simulation Conference</i>	2024
• ISE Distinguished Dissertation Award of the 2024 C.A. Anderson Awards <i>North Carolina State University</i>	2024
• Travel Award for the Annual Midwest Optimization Meeting <i>Michigan State University</i>	2023
• Mentored Teaching Fellowship <i>North Carolina State University</i>	2022
• Scholarship for Excellent Academic Records <i>Korea Aerospace University</i>	2010, 2013–2016

TEACHING EXPERIENCE

• ISE 362: Stochastic Models in Industrial Engineering <i>Instructor</i>	<i>Spring 2023</i> North Carolina State University
• ISE 441: Introduction to Simulation <i>Teaching Assistant</i>	<i>Fall 2019, Spring 2020, Fall 2020, Fall 2022</i> North Carolina State University
• ISE 748: Quality Engineering <i>Teaching Assistant</i>	<i>Spring 2019</i> North Carolina State University

- **ISE 498: Senior Design Project**

Teaching Assistant

Fall 2018
North Carolina State University

- **Analysis of Logistics System**

Teaching Assistant

Fall 2015, Fall 2016
Korea Aerospace University

RESEARCH MENTORING

- **Graduate Students**

- Qian Ni (Ph.D. Student, Civil and Environmental Engineering, Cornell University)
- Hyunwoo Shin (Ph.D. Candidate, Industrial and Systems Engineering, Virginia Tech; NREL Summer Intern)
- Hyung Khee Eun (Ph.D. Candidate, Industrial and Systems Engineering, North Carolina State University)

- **Undergraduate Students**

- Kevin Xu (Statistics, North Carolina State University)
- Wes Hankinson (Industrial and Systems Engineering, North Carolina State University)

INVITED TALKS

- **INFORMS Optimization Society Conference** 2026
A Derivative-Free Solver for Multi-Region Stochastic Equilibrium Problems
- **INFORMS Virginia Tech Student Chapter Seminar Series** 2025
Expediting Simulation Optimization
- **INFORMS Annual Meeting** 2025
Adaptive Sampling-Based Bi-Fidelity Stochastic Trust Region Method for Derivative-Free Stochastic Optimization
- **International Symposium on Mathematical Programming** 2024
Two talks: Multi-Fidelity Stochastic Trust-Region Method; Is Building First-Order Simulation Oracles Really Worth It?
- **INFORMS Optimization Society Conference** 2024
First-Order Trust-Region Methods with Adaptive Sampling
- **Annual Midwest Optimization Meeting** 2023
Common Random Numbers and Complexity in Simulation Optimization with Adaptive Sampling
- **INFORMS Annual Meeting** 2023
Consistency and Complexity of Adaptive Sampling-Based Trust-Region Optimization
- **Modeling and Optimization: Theory and Applications (MOPTA)** 2023
Expediting Stochastic Derivative-Free Optimization
- **INFORMS Annual Meeting** 2022
Complexity Analysis of Trust-Region Optimization with Adaptive Sampling for Zeroth-Order Stochastic Oracles
- **INFORMS Optimization Society Conference** 2022
Adaptive Sampling Trust-Region Optimization with Diagonal Hessian for Derivative-Free Stochastic Oracles

CONFERENCE PRESENTATIONS

- **INFORMS Simulation Society Research Workshop** 2026
Adaptive Regularization within Trust Region Methods for Stochastic Nonconvex Optimization (Poster)
- **Winter Simulation Conference** 2025
Multi-Fidelity Stochastic Trust Region Method with Adaptive Sampling
- **Institute of Industrial and Systems Engineers Annual Conference** 2025
First-Order Trust-Region Methods with Adaptive Sampling
- **Winter Simulation Conference** 2023
Towards Greener Stochastic Derivative-Free Optimization with Trust Regions and Adaptive Sampling
- **Winter Simulation Conference** 2021
Improved Complexity of Trust-Region Optimization for Zeroth-Order Stochastic Oracles with Adaptive Sampling
- **Winter Simulation Conference** 2020
Traffic Signal Control Simulation and Optimization (Poster)

ACADEMIC SERVICE

- **Journal Refereeing**

- Computational Optimization and Applications
- Journal of Scientific Computing
- Journal of Simulation
- Mathematical Programming Computation
- Numerical Algorithms
- Simulation Modelling Practice and Theory
- Statistics and Computing

- **Conference Refereeing**

- Winter Simulation Conference, 2024–2026

- **Session Chair**

- INFORMS Annual Meeting, 2022, 2023, 2025
- Modeling and Optimization: Theory and Applications (MOPTA), 2023
- Winter Simulation Conference, 2023